

Ratio Analysis

2025-12-03

Ratio Table

The data table summarizes several measles outbreaks from different states and years.

Table 1: Measles Outbreak Summary Table

Location/State	Date/Period	Total Cases	Cases_5_17	Cases_above_18	Ratio
Ohio (Amish community)	Mar-Jul 2014	383	151	163	1.079
New York City (NY)	2018-2019	649	148	122	0.824
United States	Jan-Oct 2019	1249	339	359	1.059
Texas (statewide)	Jan-Aug 2025	762	286	247	0.864
Kansas (statewide)	Jan-Aug 2025	91	35	18	0.514
New Mexico (statewide)	Feb-Aug 2025	100	23	53	2.304

Meta-Analysis of Ratios

I used a meta analysis to combine the ratios from all outbreaks into one overall estimate. Meta analysis is a statistical method that combines results from multiple datasets or studies into one pooled estimate.

Meta-analysis gives one combined ratio that:

- gives more weight to larger outbreaks
- gives less weight to smaller outbreaks
- gives a confidence interval (range of plausible values)

```
##
## Random-Effects Model (k = 6; tau^2 estimator: REML)
##
## tau^2 (estimated amount of total heterogeneity): 0.1301 (SE = 0.0982)
## tau (square root of estimated tau^2 value):      0.3607
## I^2 (total heterogeneity / total variability):    90.29%
## H^2 (total variability / sampling variability):   10.29
##
## Test for Heterogeneity:
## Q(df = 5) = 22.5633, p-val = 0.0004
##
## Model Results:
##
## estimate      se      zval      pval      ci.lb      ci.ub
## -0.0054  0.1614  -0.0332  0.9735  -0.3217  0.3110
##
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
## -----
## Random-Effects Pooled Ratio: 0.995
## 95% CI: 0.725 - 1.365
## Tau^2 (between-outbreak variance): 0.1301
## -----
```

Final result

In a random effects meta analysis of six outbreaks, the pooled 18+ to 5-17 years case ratio was **0.995 (95% CI: 0.73–1.37)**, indicating no consistent overall tendency for either adults or 5–17 year olds to have higher measles case rates on average across outbreaks. However, high heterogeneity ($I^2 = 90\%$, $\tau^2 = .13$), suggesting that the ratio varies widely between outbreaks.